

5 — Results

1 Experiments

Run matrix

Six tasks: 5-way classification, and parameter estimation for each of the five families. Each run through both arms at every feature setting.

- **PH arm**, 18 settings: 6 filtrations (Rips, alpha, DTM_{5,10,15,20}) $\times$ $\{H_0, H_1, H_0+H_1\}$
- **Classical arm**, 4 settings:
 - L on **fixed** alone (VIHRS feature set)
 - L, F, G, J all on **fixed**, stacked
 - L, F, G, J all on **sqrtn**, stacked
 - L on **fixed**, F, G, J on **sqrtn** (per-function grids, one encoder each)
- $6 \times (18 + 4) = 132$ configurations $\times$ 10 seeds = 1320 runs, all complete

Conduct

- Everything is evaluated on the same 2000 test θ per family, fixed before any run and untouched until evaluation
- Best epoch ranges 7–196 with median 34. Six runs of 1320 reached the 200-epoch cap, all of them Matérn parameters and five of them on a 1-D H_0 input
- Seed sd of a marginal cell estimate is 0.001–0.004 for accuracy, an order of magnitude below the θ bootstrap width

2 Classification

Accuracy on the whole test prior, mean over 10 seeds (chance 0.2; bootstrap half-widths $\approx$ 0.008):

	Rips	alpha	DTM ₅	DTM ₁₀	DTM ₁₅	DTM ₂₀
H_0	0.494	0.493	0.470	0.477	0.476	0.473
H_1	0.338	0.362	0.317	0.319	0.320	0.319
H_0+H_1	0.489	0.495	0.471	0.472	0.472	0.468

Classical arm:

L fixed (VIHRS set)	0.518
L, F, G, J fixed	0.539
L, F, G, J sqrtn	0.540
L, F, G, J per-function grids	0.535

Paired against the best classical setting (L, F, G, J **sqrtn**), same θ resamples:

- L alone: -0.023 $[-0.027, -0.019]$
- alpha H_0+H_1 : -0.046 $[-0.050, -0.041]$
- DTM₁₀ H_0 : -0.064 $[-0.069, -0.059]$

By regime

$\tilde{\delta}$ cells hold four families (Poisson has $\tilde{\delta} = 0$), so chance in them is 0.25; $\bar{n}$ cells hold five, so chance is 0.2.

accuracy by $\tilde{\delta}$	0.10	0.16	0.25	0.40	0.63	1.00	1.59	2.52
L, F, G, J sq rtn	0.101	0.189	0.312	0.439	0.486	0.627	0.783	0.780
L fixed	0.106	0.201	0.308	0.428	0.468	0.602	0.731	0.673
Rips H_0	0.088	0.165	0.244	0.380	0.438	0.567	0.722	0.683
alpha H_0	0.086	0.167	0.241	0.377	0.436	0.572	0.722	0.678
DTM ₁₀ H_0	0.093	0.143	0.200	0.323	0.429	0.578	0.731	0.721
DTM ₁₀ H_1	0.113	0.179	0.190	0.233	0.274	0.358	0.473	0.522

	accuracy by $\bar{n}$			recall by family (all regimes)				
	100	200	400	Pois.	Thomas	nested	Matérn	LGCP
L, F, G, J sq rtn	0.544	0.544	0.533	0.773	0.213	0.426	0.903	0.386
L fixed	0.518	0.518	0.518	0.764	0.214	0.415	0.908	0.288
Rips H_0	0.508	0.499	0.475	0.760	0.210	0.340	0.845	0.316
alpha H_0	0.509	0.498	0.472	0.758	0.200	0.339	0.848	0.321
DTM ₁₀ H_0	0.495	0.483	0.451	0.707	0.198	0.419	0.714	0.346
DTM ₁₀ H_1	0.334	0.314	0.310	0.390	0.116	0.237	0.556	0.297

Regime Breakdown

The marginal recalls above split by $\tilde{\delta}$. Restricted to one family, accuracy is that family's recall, so these are the same cells as the table above, cut by truth rather than pooled. Three settings: the best classical, the best PH (H_0 on a geometric filtration), and DTM₁₀ H_0 , the density-field filtration.

recall by $\tilde{\delta}$		0.10	0.16	0.25	0.40	0.63	1.00	1.59	2.52
Thomas	L, F, G, J sq rtn	0.027	0.033	0.062	0.080	0.156	0.286	0.466	0.615
	Rips H_0	0.043	0.048	0.070	0.106	0.190	0.273	0.424	0.536
	DTM ₁₀ H_0	0.037	0.058	0.086	0.109	0.170	0.271	0.391	0.470
nested	L, F, G, J sq rtn	0.110	0.139	0.231	0.403	0.551	0.635	0.697	0.727
	Rips H_0	0.080	0.109	0.162	0.279	0.416	0.494	0.560	0.717
	DTM ₁₀ H_0	0.116	0.145	0.219	0.356	0.532	0.653	0.672	0.746
Matérn	L, F, G, J sq rtn	0.351	0.500	0.872	0.982	1.000	1.000	1.000	1.000
	Rips H_0	0.282	0.369	0.619	0.883	0.969	1.000	1.000	1.000
	DTM ₁₀ H_0	0.210	0.270	0.312	0.537	0.706	0.863	0.986	1.000
LGCP	L, F, G, J sq rtn	0.090	0.116	0.193	0.304	0.424	0.554	0.697	0.773
	Rips H_0	0.082	0.150	0.198	0.276	0.365	0.462	0.544	0.474
	DTM ₁₀ H_0	0.089	0.115	0.198	0.290	0.408	0.496	0.556	0.666

3 Parameters

RMSE in log units, pooled over a family’s targets, over all regimes:

	Poisson	Thomas	nested	Matérn	LGCP
L, F, G, J fixed	0.063	0.878	0.593	0.118	0.627
L, F, G, J sqtrn	0.063	0.896	0.593	0.127	0.631
L fixed	0.063	0.890	0.596	0.117	0.632
Rips H_0	0.063	1.021	0.605	0.101	0.671
alpha H_0	0.063	1.020	0.605	0.102	0.672
alpha H_0+H_1	0.063	0.946	0.602	0.107	0.668
DTM ₁₀ H_0	0.063	0.945	0.606	0.158	0.682
DTM ₁₀ H_1	0.063	1.019	0.684	0.319	0.873

Regime Breakdown

Pooled log-RMSE by $\tilde{\delta}$: the best classical setting (L, F, G, J fixed) against the best PH setting, and their paired difference on the same θ resamples. *Paired* is PH minus classical on the same test θ : positive means PH is worse. * means the sign is stable, i.e. it does not flip when the test θ are resampled (95% bootstrap interval excludes 0); without it, the gap could be an accident of which θ were tested.

pooled log-RMSE by $\tilde{\delta}$		0.10	0.16	0.25	0.40	0.63	1.00	1.59	2.52
Thomas	L, F, G, J fixed	1.215	1.117	0.919	0.880	0.915	0.676	0.525	0.459
	DTM ₅ H_0+H_1	1.230	1.140	0.969	0.932	1.011	0.730	0.618	0.487
	paired	+.015	+.023	+.050*	+.052*	+.095*	+.054*	+.093*	+.028
nested	L, F, G, J fixed	0.732	0.641	0.605	0.619	0.596	0.568	0.450	0.433
	alpha H_0+H_1	0.754	0.662	0.612	0.628	0.605	0.592	0.445	0.398
	paired	+.022*	+.021*	+.008	+.009	+.010	+.024*	−.006	−.035*
Matérn	L, F, G, J fixed	0.170	0.134	0.105	0.119	0.131	0.118	0.100	0.120
	Rips H_0	0.181	0.146	0.112	0.120	0.108	0.085	0.064	0.054
	paired	+.010*	+.012*	+.007*	+.001	−.024*	−.033*	−.036*	−.066*
LGCP	L, F, G, J fixed	0.888	0.800	0.724	0.640	0.576	0.448	0.352	0.297
	alpha H_0+H_1	0.917	0.831	0.743	0.685	0.622	0.543	0.421	0.339
	paired	+.029*	+.031*	+.020*	+.045*	+.046*	+.094*	+.068*	+.042*

- Thomas, nested and LGCP improve with $\tilde{\delta}$ in both arms. LGCP improves strictly, while the two clustering families plateau at $\tilde{\delta} \approx 0.4$ – 0.6
- On Matérn the classical arm is flat over the whole range (0.105–0.170, no trend above $\tilde{\delta} \approx 0.25$) while Rips H_0 falls $0.181 \rightarrow 0.054$
- Classical arm leads everywhere except Matérn above $\tilde{\delta} \approx 0.4$ and nested’s top two bins. On nested, PH is ahead by 0.006 at $\tilde{\delta} = 1.59$ and by 0.035 at 2.52; only the second paired interval clears 0, so the top bin is the one win that is established
- Only Matérn’s gap widens with $\tilde{\delta}$. LGCP’s and Thomas’s are widest in the middle and close again at the extremes, where both arms are either uninformative or saturated.

Pooled by $\bar{n}$:

$\bar{n}$	L, F, G, J fixed			Rips H_0			DTM ₁₀ H_0		
	100	200	400	100	200	400	100	200	400
Poisson	0.079	0.061	0.043	0.079	0.062	0.043	0.080	0.062	0.043
Thomas	0.791	0.870	0.967	0.882	1.010	1.155	0.836	0.925	1.062
nested	0.520	0.586	0.660	0.521	0.602	0.678	0.534	0.605	0.668
Matérn	0.132	0.112	0.108	0.103	0.094	0.105	0.159	0.150	0.165
LGCP	0.537	0.631	0.698	0.560	0.673	0.758	0.575	0.685	0.766

- **Shape parameters get harder with $\bar{n}$** , in both arms: $\tilde{\delta}$ is defined at $\bar{n}$ and the draw is independent of it, so at fixed $\tilde{\delta}$ a larger $\bar{n}$ means a weaker per-point departure
- **Poisson is the exception.** Its target is $\bar{n}$ itself, and 0.079/0.061/0.043 tracks $1/\sqrt{\bar{n}}$ (0.084/0.059/0.042 at the bins' geometric centres) — the MLE floor for $n \mid \bar{n}$, reached by all 22 inputs

4 Observations

- PH's wins on Matérn above $\tilde{\delta} \approx 0.4$
- Classical arm wins classification overall: -0.046 paired at best, ahead in every $\tilde{\delta}$ bin from 0.16. With L alone (VIHRS) the gap is -0.023 , mostly LGCP recall ($0.288 \rightarrow 0.386$); the rest comes from F, G, J
- PH matches or beats classical on Thomas up to $\tilde{\delta} \approx 1$ and ties on Poisson. Classical wins nested against Rips H_0 at mid $\tilde{\delta}$ (0.403 vs 0.279 at 0.40), Matérn below saturation (0.872 vs 0.619 at 0.25) and LGCP at high $\tilde{\delta}$ (0.773 vs 0.474). Below $\tilde{\delta} \approx 0.6$ every method fails on Thomas (recall 0.03–0.19 vs 0.25 chance), calling 56–71% of those patterns Poisson
- H_1 adds little: at most $+0.002$ over H_0 , weakest input alone for every family and task
- Rips/alpha beat DTM on classification (0.495 vs 0.477) and Matérn; DTM wins only LGCP recall (0.386 vs 0.321), the one density-field departure. By regime the split is sharper: DTM₁₀ leads Rips on nested in every $\tilde{\delta}$ bin and on LGCP from $\tilde{\delta} \approx 0.25$ up, and trails it on Matérn in every bin below saturation
- Low- $\tilde{\delta}$ bins are below chance (0.25): the modal prediction is Poisson (68% of the bottom bin, 59% of the next) (excluded there by construction, it's a property of the prior, not the methods)
- Some targets seemingly carry no feature information: Poisson $\bar{n}$ is 0.063 for all 22 inputs ($\log n$ solves it); nested μ_1 stays at its prior sd (≈ 0.397); Thomas κ and μ agree within 0.003, being one target given $\kappa = \bar{n}/\mu$